MAIN PROJECT PROPOSAL

Adversarially Perturbed Temporal Graphs for RL-Based Detection of Multi-Agent LLM Collusion

A research proposal combining prompt-conditioned behavioural baselines, directional dependence, temporal graph structure, and reinforcement-learning-based investigation

CORE IDEA

Detect collusion from unusual directional relationships in agents' decisions, after accounting for each agent's prompt-conditioned behaviour, then use reinforcement learning to decide how and when to investigate further.

Research design draft | 24 October 2026
Status: proposed method; no experimental results claimed
Novelty status: plausible research hypothesis requiring a systematic prior-work check and empirical validation

Prepared as a research planning document. All performance claims are hypotheses until tested.

Executive summary

This project proposes a detector for collusion among LLM-based agents that does not treat similar decisions as sufficient evidence. Instead, it estimates whether one agent's history provides unusual predictive information about another agent's future actions, after conditioning on relevant task context, the target agent's own history, and its prompt condition. Those directional estimates become edges in a time-varying interaction graph. Temporal and group-level graph features then describe recurring leader-follower, reciprocal, cyclic, complementary, or coalition-like patterns. An RL investigator uses this uncertain evidence to decide which pair or subgroup to examine, whether to collect more trajectories, whether to run a permitted controlled probe, and when to alert or abstain.

The project explicitly incorporates the insight that models can exhibit characteristic behavioural tendencies, while avoiding the stronger and inaccurate claim that a model behaves the same regardless of prompt. We model an agent's action distribution as conditional on model/version, prompt, context, history, decoding settings and tools. Calibration traces estimate its normal prompt-conditioned behaviour. A controlled adversarial-instruction intervention applied to only one agent tests whether the detector mistakes an individual policy shift for collusion or instead identifies a new directional dependency between agents.

Prior work already covers information-theoretic collusion detection, steganographic coordination, RL for sequential controlled sensing, auditing LLM collusion through task regret, multi-agent activation probing, and graph-based algorithmic collusion. Therefore, neither "use graph theory" nor "use RL" is claimed as new in isolation [1–9]. The candidate contribution is the experimentally tested combination of prompt-conditioned baseline adjustment, direction-sensitive temporal dependency graphs, controlled asymmetric instruction perturbations, and an RL policy that adaptively gathers evidence under a false-alarm budget.

RESEARCH CLAIM TO TEST At a controlled false-positive rate, a prompt-conditioned directional temporal graph detector with an RL investigator may detect a broader set of collusive strategies, including complementary and asymmetric coordination, with lower observation cost or shorter detection delay than fixed-threshold and non-adaptive baselines. This is a hypothesis, not an established result.

At a glance

Item	Proposed decision
Problem	Distinguish collusion from ordinary dependence, common context, prompt-induced changes, and shared base-model tendencies.
Core evidence	Cross-validated directional predictive gain / transfer-entropy-style dependence between agent histories and partner actions.
Graph representation	Directed, weighted, time-varying graph; edges include uncertainty and prompt-condition metadata.
Adversarial component	Controlled misleading-instruction perturbations applied to one agent, both agents, or neither; optionally increase difficulty through a scripted red-team prompt policy.

Item	Proposed decision
RL role	Sequential investigator chooses to observe, inspect a pair/group, run a safe simulation probe, alert, or abstain.
Primary evaluation	False-positive-controlled detection of collusion under unilateral prompt perturbations and held-out coordination strategies.
Non-claim	The detector will not infer an agent's private intention with certainty from behaviour alone.

Contents at a glance

- 1. Motivation and precise problem statement
- 2. Prior work and research gap
- 3. Central insight: prompt-conditioned behavioural priors
- 4. Research questions and hypotheses
- 5. Threat model and operational definition of collusion
- 6. Mathematical formulation
- 7. Proposed system architecture and RL design
- 8. Experimental plan and controlled adversarial interventions
- 9. Baselines, metrics, and statistical analysis
- 10. Ablation studies and falsification criteria
- 11. Implementation plan, timeline, and resources
- 12. Risks, limitations, and responsible research
- 13. Expected contributions and novelty conditions
- 14. References

1. Motivation and precise problem statement

Multi-agent systems can produce coordinated outcomes through explicit messages, implicit response patterns, hidden communication, complementary actions, or delayed adaptation. A colluding pair need not choose the same action. One agent may deliberately take an unprofitable action to help the other, the agents may use alternating roles, or one may encode information in otherwise ordinary outputs. Conversely, two independent agents may make nearly identical decisions because they share the same state, objective, policy family, training data, or prompt. Simple action similarity conflates these cases.

Existing information-theoretic work estimates pairwise influence from game trajectories and uses mutual information and net influence to detect collusion in repeated games [1]. Recent LLM-agent work demonstrates steganographic coordination [2,3], formalizes action-based collusion using task regret and coalition advantage [4], and detects signals from model activations under distribution shift [5]. Graph-theoretic analysis of learned pricing strategies also uses policy graphs and structural measures to audit algorithmic collusion [6]. These papers make it necessary to define a specific gap rather than claim the whole problem is unstudied.

1.1 Research problem

Given multi-agent trajectories containing actions, observable messages, task state, prompt/instruction metadata, and task outcomes, estimate whether a subset of agents is coordinating in a way that violates the intended rules or objective. Do this while resisting false alarms caused by (i) shared environmental context, (ii) shared model/policy tendencies, (iii) prompt-induced changes affecting only one agent, (iv) benign adaptation, and (v) distribution shift. The detector should make decisions sequentially, because waiting for more evidence has a cost while false accusations also have a cost.

1.2 Why directional evidence matters

Pattern	Why similarity is inadequate	Evidence to seek
Complementary actions	A and B intentionally choose different but coordinated actions.	Does A's history predict B's next action beyond B's history and public context?
Asymmetric assistance	A sacrifices utility so B benefits.	Directional dependence plus payoff/objective impact; not just reciprocal similarity.
Leader-follower	B adapts after A signals or acts.	Lagged predictive dependence and temporal ordering.
Rotating support	Agents take turns helping one another.	Recurring temporal motifs and changing edge direction.
Covert signalling	A normal-looking sequence carries information used by B.	Conditional predictive information in sequences; optional internal-activation evidence where available.
Shared prompt or state	Independent agents follow the same instruction or face the same constraint.	Condition on shared state and prompt; compare against matched independent controls.

2. Prior work and the research gap

2.1 Information-theoretic game collusion

Bonjour, Aggarwal and Bhargava [1] estimate agent policies and pairwise joint policies from game records, define individual influence using mutual information, and define net influence by comparing one agent's influence on a target against the strongest alternative agent. Their experiments include iterated Rock-Paper-Scissors and Leduc Hold'em and focus on settings with two colluding agents. This is the primary statistical baseline. The present proposal does not claim that mutual information or pairwise influence is new. It asks whether directional, temporally conditioned evidence plus prompt-conditioned controls better handles asymmetric instruction perturbations and sequential LLM-agent interactions.

2.2 Hidden and steganographic collusion

Motwani et al. [2] formalize secret collusion through steganographic communication, while Mathew et al. [3] study the emergence and mitigation of steganographic collusion under misspecified incentives. These works motivate testing coordination that is not obvious from textual similarity and explain why message-content monitoring alone may be insufficient. The proposed study will use controlled synthetic

signalling strategies in a simulator and will not claim to detect every possible hidden channel from black-box actions alone.

2.3 LLM collusion audits and group-level evidence

Colosseum [4] evaluates LLM-agent collusion in cooperative task environments using objective-based regret/coalition-advantage metrics and distinguishes attempted communication from collusive actions that actually affect the task. NARCBench [5] studies activation-probe-based detection and reports generalization challenges under environment distribution shift. These works support two design decisions: label collusion using known interventions plus objective impact rather than an LLM judge alone, and separate black-box behavioural evidence from optional white-box activations.

2.4 Graph-theoretic collusion signals

Eschenbaum and Meylahn [6] construct strategy graphs for learned repeated-pricing policies and find structural metrics correlated with a profit-based collusion index. That work makes static graph structure an existing baseline, not a sufficient novelty claim. This project focuses on a different observable regime: dynamic interaction trajectories from LLM agents, with directed predictive-dependence edges, prompt interventions and sequential active investigation. The distinction must be evaluated directly; it cannot be asserted merely because the application domain differs.

2.5 RL for sequential evidence gathering

Joseph et al. [7] formulate anomaly detection with observation costs as a sequential sensing problem and use deep RL policies. Thus, applying RL to active investigation is also not new on its own. Our RL component is justified only if the state/action/reward structure uses the proposed collusion evidence and demonstrates a useful trade-off among false alarms, missed collusion, observation budget and detection delay.

2.6 Prompt sensitivity and behavioural fingerprints

Work on endpoint behavioural fingerprints shows that an LLM service's output distribution can be monitored and may vary with model version, infrastructure and inference configuration [8]. Prompt-fingerprinting work shows that prompt-conditioned token preferences can reveal model-specific signals even under transformations [9,10]. These are not evidence that an LLM has one invariant way of thinking regardless of prompt. They support a more precise design assumption: a deployed model/policy can have measurable behavioural tendencies over a controlled prompt distribution, and both the baseline and its sensitivity to prompts should be estimated.

LITERATURE-REVIEW CAVEAT This is a targeted prior-work review as of 24 October 2026, not a systematic review proving priority. Before a submission, run a reproducible search across ACL Anthology, NeurIPS proceedings, PMLR, IEEE Xplore, arXiv and relevant agent-safety venues for combinations of directional influence, temporal graphs, collusion detection, prompt perturbation, active sensing and RL.

3. Central insight: prompt-conditioned behavioural priors

The working intuition is that a model endpoint has behavioural tendencies due to its learned parameters, post-training, system configuration, decoding policy and prior conversation. These tendencies can persist across many situations, but they are not an immutable "thinking style" that is independent of the prompt. A prompt changes the conditioning information supplied to the model and can shift its action distribution. Different model versions, providers, tools and sampling settings can also change the observed policy [8].

We therefore represent an agent's policy as a conditional distribution rather than a fixed signature:

$$\pi_i(a_t \mid s_t, p_i, h_i, m_i, \theta_i, d_i)$$

where s_t is task/environment state, p_i is the current instruction or prompt condition, h_i is the agent's own interaction history, m_i identifies the model endpoint/version, θ_i describes visible decoding/configuration settings, and d_i describes available tools/permissions. The goal is to estimate normal action propensities under matched non-collusive conditions, including prompt sensitivity, then ask whether the observed inter-agent dependence exceeds what these conditions explain.

3.1 Prompt-conditioned calibration

For each model/policy family, collect independent-agent calibration episodes across clean prompts and a controlled set of benign/adversarial perturbation classes. Fit or estimate a baseline action model that predicts each agent's action from its own history, public context, prompt class/embedding, endpoint metadata and configuration. Calibration data must contain no collusive interaction. Use cross-fitting or held-out calibration episodes so a trajectory is not used both to fit and evaluate its own predicted likelihood.

3.2 The unilateral-adversary experiment

The central intervention is to apply a misleading or adversarial instruction to one agent while leaving the partner's instruction unchanged. This deliberately shifts one agent's policy distribution. The detector must not infer collusion simply because the perturbed agent becomes unusual, or because the pair's raw actions change together due to an altered task state. It should instead ask whether there is an excess directional dependency between their decisions, beyond the prompt-conditioned independent baseline, and whether the joint behaviour violates the intended objective.

Condition	Agent A instruction	Agent B instruction	What it isolates
C0: clean independent	Normal	Normal	Baseline false-positive rate.
C1: unilateral perturbation A	Misleading/adversarial	Normal	Does one-agent policy shift get mistaken for collusion?
C2: unilateral perturbation B	Normal	Misleading/adversarial	Symmetry of detector behaviour.
C3: common perturbation	Same perturbation class	Same perturbation class	Correlated changes from shared instructions.
C4: genuine collusion, clean	Normal plus scripted collusive mechanism	Normal plus scripted collusive mechanism	Detection power without perturbation.
C5: collusion plus unilateral perturbation	Perturbed or unperturbed per design	Partner remains on normal prompt	Robustness when the detector faces both phenomena.

Condition	Agent A instruction	Agent B instruction	What it isolates
C6: matched non-collusive policy shift	One independent agent changes policy for benign reasons	Normal	Negative control for adaptation or model-version effects.

3.3 Model-family controls

- Same base-model agents: tests whether shared learned tendencies create spurious dependence even when the agents do not coordinate.
- Heterogeneous model agents: tests whether model diversity changes edge estimates or false-alarm rates.
- Same model, different roles/prompts: estimates prompt/role-conditioned behaviour rather than assuming one global signature.
- Endpoint/version/decode controls: record model ID, version when available, temperature, top-p, tool set, context length, and API/provider because these can shift behaviour.
- Prompt-family holdout: test on perturbation templates not used during training/calibration to prevent prompt memorization.

4. Research questions and hypotheses

ID	Research question	Hypothesis / evidence required
RQ1	Do directional temporal dependence features detect complementary/asymmetric collusion better than symmetric similarity and mutual-information baselines?	H1: at matched false-positive rate, direction-sensitive features improve recall or detection delay on held-out leader-follower, complementary, and rotating-support strategies.
RQ2	Does prompt-conditioned calibration separate unilateral policy shifts from collusion?	H2: conditioning on prompt class, own history and context reduces false positives in C1/C2 relative to unconditioned edge scores, without erasing true collusion signals.
RQ3	Do temporal graph structures add information beyond pairwise edges?	H3: motif persistence, changes in direction, and subgroup structure improve group-level detection for scripted multi-agent coalition patterns, especially when pairwise effects are individually weak.
RQ4	Does RL-based active investigation improve the cost/accuracy trade-off?	H4: at the same false-positive budget, the RL investigator achieves lower average detection delay or lower observation/query cost than fixed thresholds and sequential-test baselines.
RQ5	Does the method generalize to unseen prompt attacks, models and environments?	H5: performance degrades less under held-out prompt families/strategies than methods trained on raw actions alone. This is a robustness hypothesis, not assumed to be true.

5. Threat model and operational definition

5.1 Observable data

The primary detector is black-box and uses only data an authorized monitor could legitimately observe: agent action records, task state, timestamps/order, messages that are part of the monitored channel, prompt/instruction metadata available to the system owner, tool-use events, reward/outcome records, and endpoint configuration metadata. It will not require private chain-of-thought. A white-box activation-probe extension is optional and must be evaluated separately because it changes the access assumptions [5].

5.2 Collusion labels

In the simulator, collusion ground truth comes from the mechanism used to generate the episode: a scripted coalition objective, a secret signalling channel, a deliberate assistance policy, or a pre-specified coordinated response rule. To avoid equating correlation with collusion, negative controls include identical policies, common instructions, common environmental shocks and unilateral prompt-induced policy shifts. Task-level harm is independently measured by comparing observed outcomes with the known intended objective or a matched independent-policy counterfactual. No single metric establishes real-world malicious intent; the study evaluates operational evidence of prohibited coordination under explicit task rules.

5.3 Adversary capabilities

- The coalition may use similar, complementary, asymmetric, delayed, alternating or steganographic action/message patterns.
- Agents may share the same base model or use different model families.
- A perturbation source may alter one agent's instructions, add misleading context, or vary the task framing while leaving others unchanged.
- The detector may see only trajectories and metadata; internal activations are excluded from the core setting.
- The RL investigator can only use observations and probes permitted by the simulated environment; active interventions are evaluation tools, not assumed available in every deployment.

6. Mathematical formulation

6.1 Notation

Symbol	Definition
A_i^t	Action taken by agent i at step t; may be categorical, structured or tool/action-event data.
H_i^t	Observed own-history of agent i through time t.
C^t	Shared task state and known common context at time t.
P_i^t	Instruction/prompt condition for agent i, including a perturbation descriptor if known.

Symbol	Definition
M_i, Θ_i	Model endpoint/version and visible inference/configuration metadata.
Y^t	Task outcome, reward, constraint violation or other outcome evidence.
K	Candidate coalition; K may contain two or more agents.

6.2 Prompt-conditioned independent baseline

Estimate the non-collusive action distribution for each agent:

$$\hat{p}_{0,i}(a \mid C^t, H_i^{t-1}, P_i^t, M_i, \Theta_i)$$

This estimates what an agent is likely to do from its own history, context, prompt condition and deployment configuration, when no coalition dependence is present. Use probabilistic models for categorical/structured actions, calibrated likelihoods, and an explicit "unknown / out-of-calibration" pathway when metadata or prompt classes are not covered. Model calibration can be a multinomial logistic model, gradient-boosted model, or a sequence model; start with the simplest calibrated baseline that is adequate for the action space.

A useful descriptive residual is the conditional surprise:

$$u_i^t = -\log \hat{p}_{0,i}(A_i^t \mid C^t, H_i^{t-1}, P_i^t, M_i, \Theta_i)$$

High surprise is a sign of an unexpected individual action, not a collusion score. It should never trigger an accusation by itself. Use it to quantify prompt-induced or policy-induced shifts and as a control feature for later relational tests.

6.3 Directed predictive dependence

For each ordered pair $i \to j$, compare a target-agent model that sees only j's history and context with a model that additionally sees i's history. Evaluate log scores on held-out, time-respecting folds:

$$g_{i \to j} = (1/N) \, \Sigma_t \left[\log \hat{p}_1(A_j^{t+1} \mid H_j^t, X^t, H_i^t) - \log \hat{p}_0(A_j^{t+1} \mid H_j^t, X^t) \right]$$

Here X^t collects observed shared context and all available exogenous prompt/perturbation assignments, model endpoint/version metadata, and decoding/configuration settings for both agents. Both models receive the same X^t; only the expanded model receives the source agent's behavioural history H_i^t. This prevents the edge from receiving credit merely because the source prompt or model identity is predictive. The held-out gain measures the additional predictive value of i's history for j's next action. Under correctly specified distributions and adequate data, its population analogue is connected to conditional mutual information / transfer entropy [11]. In finite samples it remains model-dependent, so use blocked cross-validation and uncertainty intervals rather than treating it as exact information-theoretic truth.

A transfer-entropy-style definition for the same directional question is:

$$T_{i \to j} = I(H_i^t \, ; \, A_j^{t+1} \mid H_j^t, X^t)$$

Here X^t is the shared set of observed context and exogenous prompt/model/configuration variables defined above. This quantity is non-negative and does not, by itself, reveal whether the relationship is helpful or harmful. It also does not establish causal influence: a common unobserved factor may affect

both agents. If messages are visible and are part of the causal pathway being audited, represent them explicitly with timestamps and sensitivity analyses; do not accidentally condition away the very signal the detector is designed to find. Variables that are mediators of the relationship being tested should not automatically be conditioned out; report the conditioning set explicitly.

6.4 Directional asymmetry and dependence strength

$$D_{ij} = g_{i\to j} - g_{j\to i}$$

$$S_{ij} = \max(0, g_{i\to j}) + \max(0, g_{j\to i})$$

$$A_{ij} = (g_{i\to j} - g_{j\to i}) / (|g_{i\to j}| + |g_{j\to i}| + \varepsilon)$$

D describes signed asymmetry; S describes total positive predictive gain; A is a bounded descriptive asymmetry index when its denominator is nonzero. None is a collusion label. A high D can represent a benign leader-follower relationship, and a near-zero D can represent reciprocal coordination or no relationship at all. Report strength, direction, uncertainty and outcome impact together.

6.5 Conditional null test and robust calibration

For each ordered edge, generate a conditional null distribution by permuting or resampling the source history in blocks within matched context/prompt strata, preserving temporal structure as far as the design requires. Compute a one-sided or two-sided p-value based on the pre-registered test. Across many ordered edges, control multiplicity using Benjamini-Hochberg or a family-wise method selected in advance. Also report effect sizes and confidence intervals; a p-value alone is not a practical effect size.

As a secondary scaling step, robustly normalize an edge statistic against an independent-agent reference distribution:

$$z_{i\to j} = [g_{i\to j} - \text{median}(G_0)] / [1.4826 \times \text{MAD}(G_0)]$$

G_0 is the distribution of the same statistic under matched non-colluding controls. The factor 1.4826 is the conventional asymptotic normal-consistency scaling for MAD; it is not a magic correction and should not be used when MAD is zero or the baseline is poorly sampled. Use a fallback scale estimate and report raw effect sizes as well. Normalization supplements rather than replaces the null test.

6.6 Prompt-shift contrast

For unilateral prompt perturbations, estimate the difference between the directional edge statistic in the perturbed and matched clean conditions:

$$\Delta g_{i\to j}\hat{}(P_k) = g_{i\to j} \mid P_k \text{ perturbed} - g_{i\to j} \mid \text{matched clean}$$

Interpret this contrast through the factorial control design, not in isolation. If perturbing agent i shifts i's own action distribution but does not create excess dependence with j, the detector should treat it as an individual policy shift. If the perturbation creates apparent dependence through a shared environment change, the common-perturbation control should reveal the confound. A genuine coordinated mechanism is labelled independently in the simulator so sensitivity and specificity can be measured.

6.7 Time-varying directed graph

$$G_t = (V, E_t, W_t, U_t)$$

V is the agent set; E_t contains candidate directed edges; W_t stores effect-size weights (e.g., $g_{i\to j}$ or its calibrated z-score); U_t stores uncertainty, sample count, prompt condition and calibration status. Edges must not be interpreted as causal merely because they are directed. The graph is updated over time windows or by an online estimator, with minimum sample requirements to avoid unstable edges from short traces.

6.8 Temporal and group-level features

Feature family	Examples	Purpose / caution
Edge persistence	Proportion of windows with significant excess dependence; run length of edge activity.	Distinguishes transient coincidences from recurring structure; persistence alone is not intent.
Directional motifs	A→B followed by B→C; recurring A→B→A response; cycles over time.	Captures multi-step coordination and rotating support; compare to matched null graphs.
Reciprocity and asymmetry	Both directions significant; D_{ij} and S_{ij}.	Separate strong reciprocal dependence from leader-follower-like patterns.
Subgroup structure	Directed communities, density relative to outside edges, group-level dependence.	Find coalitions larger than pairs; control false discovery across groups.
Change points	Sudden shift in edges, motif counts or dependence profile.	Detect onset/cessation of coordination; benign prompt changes also cause change points.
Outcome linkage	Task regret, coalition advantage, protocol violation, harm to non-coalition agents.	Provides objective evidence beyond relationship structure; use known simulator objective/counterfactual.

Graph motifs and community scores will be compared against degree-preserving or condition-matched null graphs so that large groups, dense interaction patterns, or high activity do not look suspicious merely because of network size. Static strategy-graph metrics from [6] are included as related baselines where their required policy access is available.

7. RL investigator: sequential decision-making layer

7.1 Why RL is included

The RL agent is an investigator, not a colluding player. It receives uncertain, time-varying statistical evidence and chooses whether to observe more, inspect a relationship, run a controlled probe in the simulator, or issue an alert. The decision has sequential trade-offs: more data may reduce uncertainty, but observations cost time/API calls and false allegations are costly. RL for sequential controlled sensing already exists [7]; this project must show that the learned policy is useful for this particular graph-based collusion-audit setting.

7.2 POMDP definition

Component	Definition
Hidden state	True coalition membership/mechanism, latent policy state, whether coordination is active, and any unavailable internal state.

Component	Definition
Observed state o_t	Graph edge weights and uncertainty, motif counts, prompt-shift estimates, individual-surprise features, objective/regret signals, elapsed steps, observation budget, and model/prompt metadata.
Actions	Continue observation; inspect a candidate pair; inspect a subgroup; request additional allowed logs; run a controlled simulation probe; flag pair/group; declare no evidence / abstain.
Transition	New agent actions, messages, environment state and task outcomes arrive; graph and evidence estimates update.
Reward	Reward correct, timely detection; penalize false positives, false negatives and unnecessary observation/probe cost. Weights reflect the intended deployment trade-off and must be sensitivity-tested.
Policy	A recurrent PPO investigator over engineered graph/statistical features; compare against DQN or a simpler policy if appropriate. A GNN encoder is an extension/ablation, not required to make the primary hypothesis testable.

7.3 Reward design

$$r_t = -c_obs \cdot I[observe] - c_probe \cdot I[probe] + r_terminal$$

$$r_terminal = R_TP \cdot I[correct\ alert] + R_TN \cdot I[correct\ no\text{-}alert] - C_FP \cdot I[false\ alert] - C_FN \cdot I[missed\ coalition] - C_A \cdot I[unresolved\ termination]$$

Set C_FP, C_FN, R_TP, R_TN, c_obs and c_probe in advance for each target operating regime. Run reward-weight sensitivity analyses because arbitrary reward ratios can manufacture a preferred policy. Training reward should not directly expose a hidden label at decision time; in simulation, terminal feedback may be used for training, while the evaluation report must state exactly what feedback is available during training and what is observable at deployment. Consider separate policies for high-stakes conservative auditing versus low-cost screening.

7.4 RL baselines and fair comparison

- Fixed threshold on the original information-theoretic net-influence score.
- Threshold on directional edge statistics with prompt-conditioned null calibration.
- Sequential probability ratio / Bayesian evidence threshold with a predefined stopping rule.
- A non-RL active investigator using uncertainty or expected information gain to choose the next pair/probe.
- The same graph encoder with a supervised classifier if labels are available, to test whether sequential RL is actually needed.
- RL without prompt-conditioned features, without temporal graph features, and without outcome/regret features as ablations.

IMPORTANT COMPARISON RULE Compare all methods at matched false-positive budgets and observation availability. If RL uses more observations or gets privileged simulator information unavailable to a baseline, the comparison is not fair.

8. Proposed end-to-end algorithm

1. **Step 1 - Calibrate:** Collect independent-agent trajectories under normal prompts, matched task contexts, benign variations and controlled perturbation classes. Fit held-out action-propensity models and document endpoint/configuration metadata.
2. **Step 2 - Run task episodes:** Instantiate a known task environment with scripted independent or collusive mechanisms; record actions, messages, context, timestamps, prompts, outcomes, and ground-truth mechanism.
3. **Step 3 - Estimate individual shifts:** Compute conditional surprise and prompt-shift diagnostics. Mark out-of-calibration conditions instead of overconfidently extrapolating.
4. **Step 4 - Estimate directional dependence:** For every ordered pair, compute cross-validated predictive gain and/or a transfer-entropy-style estimate, with block-permutation null tests.
5. **Step 5 - Update graph:** Build the time-varying directed graph with effect sizes, significance/uncertainty, sample counts and condition metadata; correct for multiple testing.
6. **Step 6 - Compute structural evidence:** Update motif persistence, directional asymmetry, community structure and change-point features; calculate objective harm/regret separately.
7. **Step 7 - RL investigation:** Provide the observed evidence to the investigator. It selects additional observations, pair/subgroup checks, allowed probes, alert, or abstention.
8. **Step 8 - Record evidence:** Log which evidence caused the alert, uncertainty, condition, observation cost, and the limits of the conclusion. Produce an auditable result rather than a bare black-box probability.
9. **Step 9 - Evaluate:** Compare with baselines on unseen seeds, strategies, prompt perturbations and model combinations; estimate confidence intervals and run ablations.

9. Experimental plan

9.1 Environments

Primary environment: a small cooperative task-allocation/scheduling problem represented as a distributed constraint optimization problem (DCOP), following the kind of verifiable task-outcome design used by Colosseum [4]. The intended objective is explicit, such as minimizing total completion cost while satisfying constraints. This gives a defensible way to measure task regret and identify coalition-induced objective degradation.

Secondary controlled environment: a repeated multi-agent strategic game with discrete actions and controlled observability. This is useful for sanity-checking directional dependence estimates and connecting directly to the classic benchmark family in [1]. Start with three or four agents; expand to larger groups only after the detector and evaluation pipeline work.

9.2 Agent and coalition mechanisms

Mechanism	Ground-truth construction	Signature expected but not assumed
Independent baseline	Agents receive individual policies and no coordination mechanism.	No excess dependence after matched context/prompt conditioning, except sampling noise.

Mechanism	Ground-truth construction	Signature expected but not assumed
Shared-model independent	Agents use the same underlying model and role template but no communication/coordination channel.	Tests common model tendencies as a false-positive confound.
Complementary coalition	Agents choose different actions according to a scripted joint objective.	Raw action similarity may be low while conditional predictive dependence is high.
Leader-follower	A's actions or signals systematically change B's response policy.	Asymmetric lagged dependence.
Reciprocal / alternating support	Agents take turns making concessions or assistance decisions.	Temporal reciprocity or recurring motifs.
Hidden-channel / encoded signal	A simulator-provided channel conveys a pre-defined signal that changes a partner's later action.	The detector should test temporal information transfer; text-only similarity may fail.
Benign adaptation	Agents adapt to public state or partner actions to improve the legitimate task.	Tests whether dependence alone over-accuses cooperation.

9.3 Adversarial perturbation protocol

Perturbations should be defined as a controlled experimental factor, not an untracked collection of prompt hacks. Prepare a taxonomy: (i) misleading task detail; (ii) conflicting local instruction; (iii) instruction phrased to encourage a strategy shift without changing the shared state; (iv) role framing/format change; and (v) held-out paraphrases. Use low/medium/high severities with documented intended effects. Avoid using perturbation templates in both train and test. The test asks whether the detector separates an individual response shift from increased inter-agent dependence and prohibited objective impact.

- Apply the perturbation to A only, B only, both, or neither in a randomized factorial design.
- Cross each condition with independent, benignly adaptive, and collusive agent mechanisms.
- Include a manipulation check: confirm that the perturbation measurably shifts the perturbed agent's action distribution; otherwise it was not a meaningful treatment.
- Include a common-cause control: vary public task state or shared prompt in ways that shift both agents without collusion.
- Use held-out perturbation families for final robustness testing and report severity-conditioned performance.

9.4 Models, repetitions and data split

Use at least two or three model families or policy implementations if resource access permits, with pinned version/configuration metadata. Include same-model pairs and heterogeneous pairs. Start with deterministic scripted agents to validate ground-truth and statistical calculations, then add LLM agents. Repeat every stochastic condition across independent episode seeds. A target of approximately 100 independent episode seeds per key confirmatory condition is a reasonable planning goal, but the final number should be set using pilot variance and a pre-registered power or precision calculation; do not present the planning target as a guarantee of adequate power.

Split data by episode seed, prompt family, collusion policy/mechanism and, where feasible, model family. Never split adjacent windows from a single episode across train and test. Reserve an out-of-

distribution test where the coordination mechanism and perturbation template are unseen during training. Tune thresholds and reward weights using validation data only.

10. Baselines and evaluation

10.1 Baseline methods

Baseline	Why it is required
B1: original information-theoretic detector [1]	Direct prior-work comparator; implement its influence/net-influence threshold faithfully.
B2: symmetric conditional dependence / mutual information	Tests whether directionality adds value over a symmetric dependence measure.
B3: directional predictive gain / transfer entropy only	Tests the core edge statistic without graph motifs or RL.
B4: static strategy-graph metrics [6]	Tests if dynamic interaction graphs add value beyond static policy graph structure, where comparable access exists.
B5: text similarity / action similarity / LLM judge	Weak but relevant practical comparators; must not be presented as statistically equivalent to the stronger baselines.
B6: task regret / coalition advantage [4]	Outcome-based complementary detector; useful for checking whether relationship scores align with actual objective harm.
B7: activation probes [5], if white-box access exists	Separate white-box comparison; not a fair black-box baseline if the main system cannot access activations.
B8: sequential hypothesis test / active sensing [7]	Tests whether RL improves the investigation policy beyond established sequential evidence gathering.
B9: supervised classifier on the same features	Tests whether RL contributes beyond an ordinary classifier when labels are available.

10.2 Metrics

Metric	Definition / purpose
AUROC and AUPRC	Overall ranking quality; AUPRC is especially important with rare collusion.
FPR at fixed TPR / TPR at fixed FPR	Primary safety comparison because false accusation rate is a central operational constraint.
Precision, recall, F1	Threshold-specific summary; always report threshold and prevalence.
Detection delay	Steps or observations from onset of collusion to the first correct alert.
Observation / query cost	Number of additional trajectories, probes or API actions requested by the RL investigator.
Calibration	Brier score, reliability diagrams or expected calibration error for collusion probabilities where used.
Robustness gap	Performance drop from clean conditions to unilateral perturbation and held-out strategy/model conditions.
Group localization	Pair/group precision and recall; exact coalition recovery for scripted groups.

Metric	Definition / purpose
Task impact / regret	Difference between observed objective and intended optimum / matched independent baseline.

10.3 Statistical analysis

- Use episode-level confidence intervals; do not treat every action in a long trajectory as an independent sample.
- Bootstrap over independent episodes/seeds, and use paired comparisons where methods share the same test episodes.
- Use blocked conditional permutations for edge significance and predefine the blocking/stratification variables.
- Correct for multiple comparisons across ordered agent pairs and candidate subgroups.
- Pre-register primary metrics, operating points, exclusion rules, model versions, prompt families and confirmatory hypotheses.
- Report effect sizes and uncertainty, including negative results. Do not infer causality from predictive edges alone.

11. Ablations and falsification criteria

The project should have explicit kill criteria. If the results do not support the novelty hypothesis, the conclusion should reflect that instead of adding complexity until a positive result appears.

Ablation / test	Question	Falsification or redesign signal
Remove prompt-conditioned baseline	Does calibration help separate individual shifts from collusion?	No improvement in unilateral-perturbation false positives, or it destroys recall.
Replace directional edges with symmetric MI	Does direction matter?	No benefit on complementary/asymmetric held-out strategies at matched FPR.
Remove temporal motifs	Do recurring graph patterns add information?	No generalization/recall gain beyond pairwise edge statistics.
Remove objective/regret features	Does task harm evidence help distinguish benign dependence?	No value over dependence alone, or task impact does not track the label definitions.
Replace RL by a fixed threshold / SPRT / active sensing heuristic	Does RL help?	No Pareto improvement in delay, cost and false alarms. If so, retain the graph detector but drop claims of RL advantage.
Hold out prompt family and model family	Does the method generalize?	Performance collapses outside training conditions; narrow the claim or redesign calibration.
Same-model independent negative control	Does shared base-model behaviour cause false alarms?	High FPR means the baseline does not control common policy tendencies adequately.
Perturb agent without partner interaction	Does the method mistake odd behaviour for collusion?	FPR rises sharply with only the perturbed agent changed; central robustness claim fails.

12. Implementation plan and timeline

Phase	Work packages	Deliverable / exit criterion
Phase 1 (Weeks 1-2)	Reproduce baseline paper calculations on a small simulator; define collusion scripts and logging schema; unit-test information metrics.	Reproducible baseline implementation; verified labelled episodes; documented equations and metric tests.
Phase 2 (Weeks 3-4)	Implement prompt-conditioned calibration; directional predictive gain/transfer entropy; conditional permutation tests; temporal graph.	Edge estimates validated on synthetic dependencies and independent controls.
Phase 3 (Weeks 5-6)	Build controlled prompt-perturbation matrix; run baseline comparisons; implement task-level regret/coalition metrics.	Pilot result on unilateral prompt perturbation and complementary collusion; revised sample-size plan.
Phase 4 (Weeks 7-8)	Train RL investigator; evaluate versus SPRT, thresholds and active sensing; run ablations, held-out tests, and error analysis.	Final benchmark report, code, configs, seeds, plots and a paper-ready results table.

12.1 Practical stack

- Python for simulation, statistical estimation and evaluation; NumPy/SciPy/scikit-learn for statistical models and tests.
- NetworkX for a first implementation of dynamic directed graph features; introduce a GNN only if the non-neural features prove insufficient.
- Gymnasium-compatible environment for sequential investigation and an established RL library for PPO; pin versions and log seeds.
- Local/open-weight LLMs or permitted API endpoints, with versions/settings recorded. Never rely on undocumented provider defaults for reproducibility.
- JSONL or Parquet trajectory logs with one row per action/event and explicit agent, timestamp, prompt-condition and episode IDs.

13. Risks, limitations, and responsible research

Risk	Mitigation
False positives from common context	Condition on observed task state; include shared-prompt and shared-model controls; calibrate edge nulls.
Prompt perturbation changes task semantics	Use a factorial design and separate prompt-induced policy shift from objective-level collusion; document exactly what the perturbed instruction changes.
LLM nondeterminism and endpoint drift	Repeat episodes, pin endpoint/config metadata, monitor calibration drift, and add uncertainty/out-of-distribution handling [8].
Unobserved common causes	Describe edges as predictive dependence, not causal edges; use controlled interventions where feasible.
Hidden signalling evades black-box analysis	Limit claims to observable channels; report failure cases; optionally compare to white-box probes [2,3,5].
RL overfits scripted mechanisms	Hold out prompt templates, model families, coalition policies and environment variants; include non-RL baselines.

Risk	Mitigation
Reward design creates the desired result	Predefine reward ratios, perform sensitivity analysis, report Pareto curves and compare multiple cost regimes.
False claim of intent	Use “suspected collusion” or “evidence of prohibited coordination”; do not equate correlation or objective harm with private intent automatically.

Only use controlled simulated tasks and authorized model endpoints. The goal is detection evaluation, not building a covert communication channel for deployment. If secret signalling is simulated, use a closed toy protocol and publish enough detail for reproducibility without turning the work into a recipe for evading real monitoring. Avoid logging sensitive real-world user data.

14. Expected contributions and novelty conditions

If the hypotheses are supported, the project could contribute:

- A prompt-conditioned estimation protocol that explicitly separates an individual agent’s response shift from relational evidence.
- A directional, time-varying graph representation of inter-agent predictive dependence with calibrated uncertainty.
- A benchmark protocol that combines unilateral adversarial instruction perturbations with known collusion mechanisms and benign negative controls.
- An RL investigator that chooses what evidence to gather and when to stop, evaluated against fixed-threshold and established sequential-testing baselines.
- An empirical analysis of which mathematical signal families generalize to complementary, asymmetric, hidden-signal and group-level coordination.

NOVELTY CLAIM THAT IS DEFENSIBLE BEFORE RESULTS “We propose and evaluate a prompt-conditioned, direction-sensitive temporal-graph approach for collusion auditing, paired with RL-based active investigation, and study whether it reduces false alarms under unilateral instruction perturbations.” The stronger claim that it improves detection must await experiments; the claim that it is the first of its kind requires a broader, documented systematic search.

14.1 Decision rules for the project

- If directional features do not beat symmetric MI on complementary/asymmetric strategies, do not claim directional novelty improves detection.
- If prompt-conditioned calibration does not reduce false positives under unilateral perturbations, revisit the baseline model and confound controls.
- If RL does not improve cost-delay-accuracy trade-offs at a matched false-positive rate, retain it only as a negative result or remove it from the final system.
- If generalization fails across prompt families or model families, narrow the claim to the tested regime and report the failure explicitly.

15. References and source notes

Reference numbering used throughout this proposal is listed below. Dates/venue labels follow the current publisher or preprint records checked for this draft. The list is a targeted starting point and not a claim of exhaustive coverage.

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